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Jan 25, 2025
Mobile Robotics
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Write-Up

What I did:

I implemented a single ROS2 node, called "Driver", which sensed obstacles, planned an avoidance course, and sent twists to the Stage cmd_vel topic. The obstacle avoidance algorithm I implemented was the Vector Field Histogram. This algorithm uses lidar to sense distances from the robot, find directions free of near obstacles, and returns the angle direction most closely aligned with the target angle. After the angle of action was decided, a proportional controller based on distance to target error was used to control speed. The proportional value was discovered through testing.

Where it worked:

I found the Vector Field Histogram to be surprisingly robust. In most cases the algorithm successfully navigated many complicated obstacles and walls in search of the target. However, the algorithm does have some weaknesses. The algorithm takes greedy actions with respect to open angles, allowing it to get trapped in local minimas, failing to converge to a solution. The algorithm also requires parameter tuning which can be challenging in and of its own – my implementation struggled with tight turns. If paired with a navigation algorithm, I believe it would be incredibly robust.

What I learned about avoiding obstacles in both a simulated and real robots:

I learned that avoiding obstacles is a challenging task. I learned a few of the algorithms that exist at a high level, and the vector field histogram at a low level. Through this project's experience, I discovered the difficulty of parameter tuning, and the trade offs that must be weighed and managed. It also became blaringly obvious the challenges that exist between converting simulated results into the real world. Simulation provides a simple, predictable environment for the agent to act in, but the real-world contains many dynamic obstacles making it hard.